Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/14/2007
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Analysis of a Novel Sensory Mechanism in Root Phototropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received extension to 07/31/2009 per A. Chu (6/08) NOTE: Received extension to 07/31/2008 per A. Chu (5/07) Note anniversary date changed to 8/1; start date remains the same (per A. Chu 4-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NCC2-1200		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. The phototropic response in roots is much weaker than the graviresponse, which competes with and often masks the phototropic response. It was through the use of mutant plants with a weakened gravitropism that we were able to identify the activity of the red-light-dependent phototropic system. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. In this project, we plan to use the European Modular Cultivation System (EMCS) on the ISS. We are working toward a launch date of July 2006 on STS-121 for the EMCS and our experiments. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA. Our spaceflight project is termed TROPI for tropisms and used the European Modular Cultivation System (EMCS) onboard the ISS. TROPI was performed in October and December 2006 on the ISS. Some samples were returned on STS-116 in December 2006 and the balance of samples will return in June 2007 on STS-117 and/or on a later shuttle mission in the second half of 2007. Past year's Accomplishments--Research/Development Activities and Spaceflight Support 1. Biocompatibility test at ARC. Since the launch of STS-121 was postponed to July 2006, we did some further biocompatibility tests with seeds and EUE at ARC, 2. Flight build at ARC. The launch date of STS-121 was changed to July 2006. Thus, we had a complete flight build of 24 EC's in March-April 2006. The post-doc and ARC staff performed the final hardware build under careful QA supervision. 3. Increment 14 science symposium at JSC. There was a symposium for all scientific experiments on Increment 14 of the ISS at JSC. The PI gave a presentation and science overview at this symposium in June 2006. 4. Hardware handover at KSC. Based on the launch date of July 2006, the PI and ARC staff had a handover of the hardware at KSC in mid-June 2006. We were limited to 8 of the 24 EC's. 5. July 4 launch of STS-121. Eight TROPI EC's were launched on STS-121. The PI and co-PI were at KSC in case there were last minute issues with any aspect of TROPI. 6. September 9 launch of STS-115. The balance of 16 EC's was launched on STS-115. The PI was at KSC during both launch attempts (August and September) and had to deal with a number of technical issues during the rollback of STS-115. 7. Flight operations support at N-USOC in Trondheim, Norway. EMCS flight operations occurred at N-USOC in Trondheim, Norway. TROPI consisted of three six-day runs. During TROPI operations, the PI and ARC team had to monitor video downlinks and had to make a number of decisions on off-nominal events (e.g. power outages, incorrect command sequences up-linked, computer freezes, etc.). During run 1 in October 2006, the PI supported operations in Trondheim, and the co-PI supported run 2 in Trondheim in late October 2006. 8. ASGSB meeting in Arlington VA. In November 2006, the PI and his team gave several presentations on the EMCS project and related ground-based research at the 2006 ASGSB meeting. This is an important international meeting for researchers with spaceflight experiments. 9. Landing of STS-116 on December 22, 2007 at KSC. STS-116 returned frozen samples from 1 run, and videotapes from 2 runs. The PI was at KSC to transfer frozen samples from JSC cold bags to a dry nitrogen dewar (for transfer to University of Florida). 10. Flight operations support at N-USOC in Trondheim, Norway. Dr. Melanie Correll was at N-USOC to support flight operations during run 3 of TROPI. 11. Retrieval of videotapes at KSC. The hi-8 video tapes were returned to the PI team in early February 2007, and the co-PI traveled to KSC to personally receive the tapes for the PI group. 12. Video image analysis at Miami University. Once the PI team obtained the tapes, germination, growth, and curvature data will be obtained from images recorded during the flight. The tapes first need to be digitized since we will work with digital images and the digitization is in progress. This process is very time consuming due to the large number of background video tape images without true data images. Growth and curvature will be measured from images digitally captured with the image analysis program Image-Pro Plus on a PC computer. 13. Analysis of frozen samples at University of Florida. One of the goals of this project is to study gene expression as

	the result of light and gravity-treatments of plants in space. As a first step in this process, RNA needs to be isolated from the frozen plant samples. We are in the process of analysis of RNA and will short begin microarray experiments. 14. Throughout the funding year. There were numerous telecons among the PI team at Miami University and the ARC team. Detailed minutes from each telecon are available upon request. Some highlights: • The final version of the EUE has been developed and tested. Numerous spaceflight experiment feasibility test have been performed.(e.g. seed viability, long term storage, etc.) • We have continued microarray experiments on root phototropism on both WT and phy strains of Arabidopsis. • Post-flight procedures, including handling the frozen specimens, have been finalized. • Numerous videocamera and programming tests of the EMCS have been performed. • The computer-based system to convert the images from the video tape to digital images has been developed and tested. • We have built up seed stock (with high germination rates) for the spaceflight experiments. At least two generations of seeds will be available of the wild-type and mutant strains of Arabidopsis.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Kumar P, Correll MJ, Mullen JL, Hangarter RP, Edelmann RE. "Phototropism in space. " American Society of Plant Biologists annual meeting, Chicago, IL, July 7-11, 2007. American Society of Plant Biologists annual meeting. Abstract Book. 2007. , Jul-2007
Abstracts for Journals and Proceedings	Molas ML, Palmieri M, Kiss JZ. "PKS1 and PHYA are involved in red-light-based positive phototropism in roots." American Society of Plant Biologists annual meeting, Chicago, IL, July 7-11, 2007. American Society of Plant Biologists annual meeting. Abstract book, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Montgomery CE, Kumar P, Kiss JZ. "The role of phytochrome C in gravitropism and phototropism." American Society of Plant Biologists Midwestern Section annual meeting, E. Lansing, MI, March, 2007. American Society of Plant Biologists Midwestern Section annual meeting. Abstract Book, p. 3, 2007. , Mar-2007
Abstracts for Journals and Proceedings	Kuntz AL, Molas ML, Kiss JZ. "Examining biological variability: a microarray study." 22nd Annual Meeting of the American Society for Gravitational and Space Biology, Arlington, VA, November 2006. Gravitational and Space Biology 2006 Nov;2(1):4. , Nov-2006
Abstracts for Journals and Proceedings	Cirelli D, Kiss JZ. "The effect of microgravity on the arrangement of amyloplasts in hypocotyl endodermal cells." 22nd Annual Meeting of the American Society for Gravitational and Space Biology, Arlington, VA, November 2006. Gravitational and Space Biology 2006 Nov;20(1):18. , Nov-2006
Abstracts for Journals and Proceedings	Molas ML, Correll MJ, Kiss JZ. "Positive phototropism in roots: who is involved?" 22nd Annual Meeting of the American Society for Gravitational and Space Biology, Arlington, VA, November 2006. Gravitational and Space Biology 2006 Nov;20(1):10. , Nov-2006
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Abstracts for Journals and Proceedings	Kiss JZ, Kumar P, Molas ML, Correll MJ, Bowman RN, Eodice MT, Edelmann RE. "Development of experiments on the interaction between gravitropism and phototropism in Arabidopsis for spaceflight studies." Committee on Space Research (COSPAR), 36th Scientific Assembly, Beijing, China, July 2006. COSPAR Abstract Book, 2006. , Jul-2006
Abstracts for Journals and Proceedings	Molas ML, Correll MJ, Stevens MH, Kiss JZ. "Gene profiling studies of red-light-induced signal transduction in roots. " American Society of Plant Biologists Annual Meeting, Boston, MA, August 2006. American Society of Plant Biologists Annual Meeting, Abstract Book, p. 258, August 2006. , Aug-2006
Abstracts for Journals and Proceedings	Palmieri M, Duley SM, Schwind MA, Kiss JZ. American Society of Plant Biologists, Boston, MA. Abstract Book "Myosin plays a role in gravitropism in Arabidopsis hypocotyls. " American Society of Plant Biologists Annual Meeting, Boston, MA, August 2006. American Society of Plant Biologists Annual Meeting, Abstract Book, August 2006. , Aug-2006
Articles in Peer-reviewed Journals	Molas ML, Kiss JZ, Correll MJ. "Gene profiling of the red light signalling pathways in roots." J Exp Bot. 2006;57(12):3217-29. Epub 2006 Aug 14. PMID: 16908503 , Sep-2006
Articles in Peer-reviewed Journals	Kumar P, Kiss JZ. "Modulation of phototropism by phytochrome E and attenuation of gravitropism by phytochromes B and E in inflorescence stems." Physiologia Plantarum, 2006 Jun;127(2):304-11. http://dx.doi.org/10.1111/j.1399-3054.2006.00670.x , Jun-2006
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Articles in Peer-reviewed Journals	Vitha S, Yang M, Sack FD, Kiss JZ. "Gravitropism in the starch excess mutant of Arabidopsis thaliana." American Journal of Botany 2007 Apr;94(4):590-8. , Apr-2007

Articles in Peer-reviewed Journals	Kiss JZ, Kumar P, Bowman RN, Steele MK, Eodice MT, Correll MJ, Edelmann RE. "Biocompatibility studies in preparation for a spaceflight experiment on plant tropisms (TROPI)." Adv Space Res. 2007;39(7):1154-60. http://dx.doi.org/10.1016/j.asr.2006.12.017 , Jun-2007